

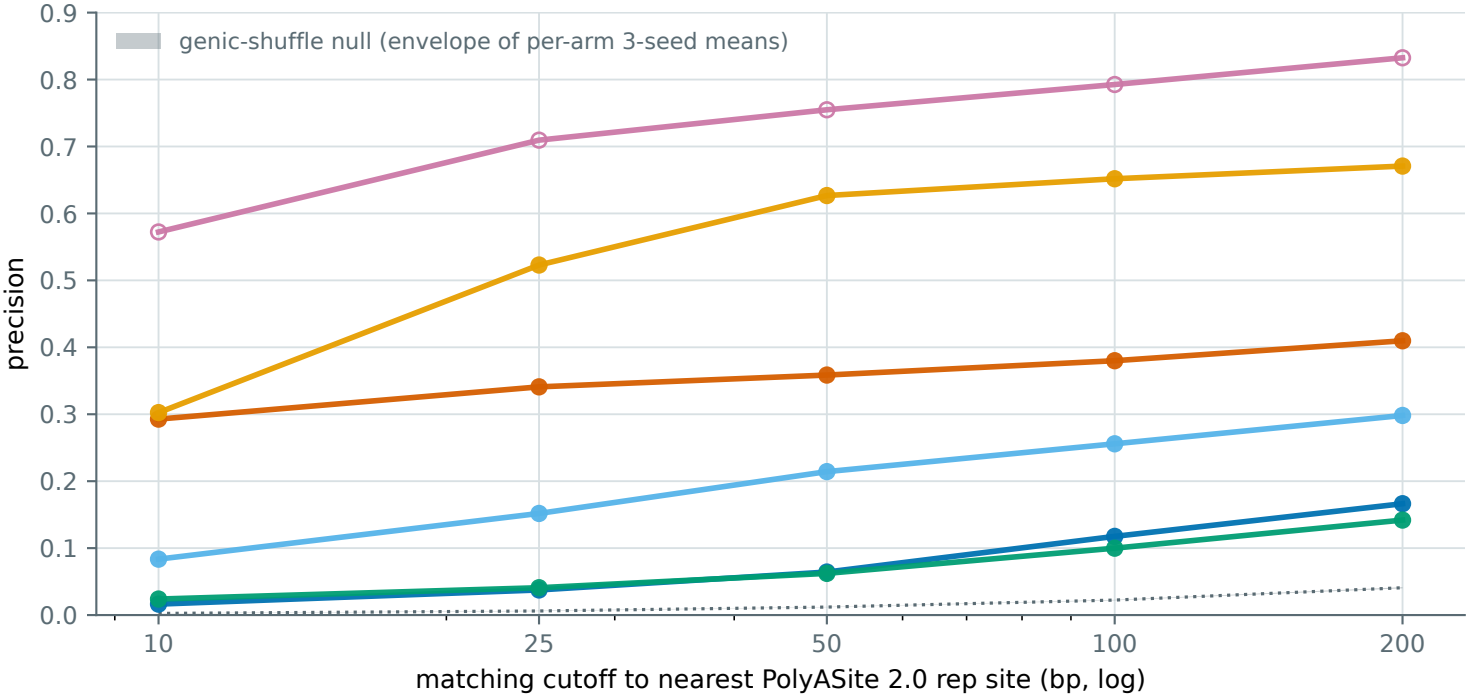
Six PAS callers, two datasets: a trade surface, not a ranking

PeakATail ranks LAST of five de novo tools on human PBMC (F1 0.134) and second on mouse testis (0.258-0.260)

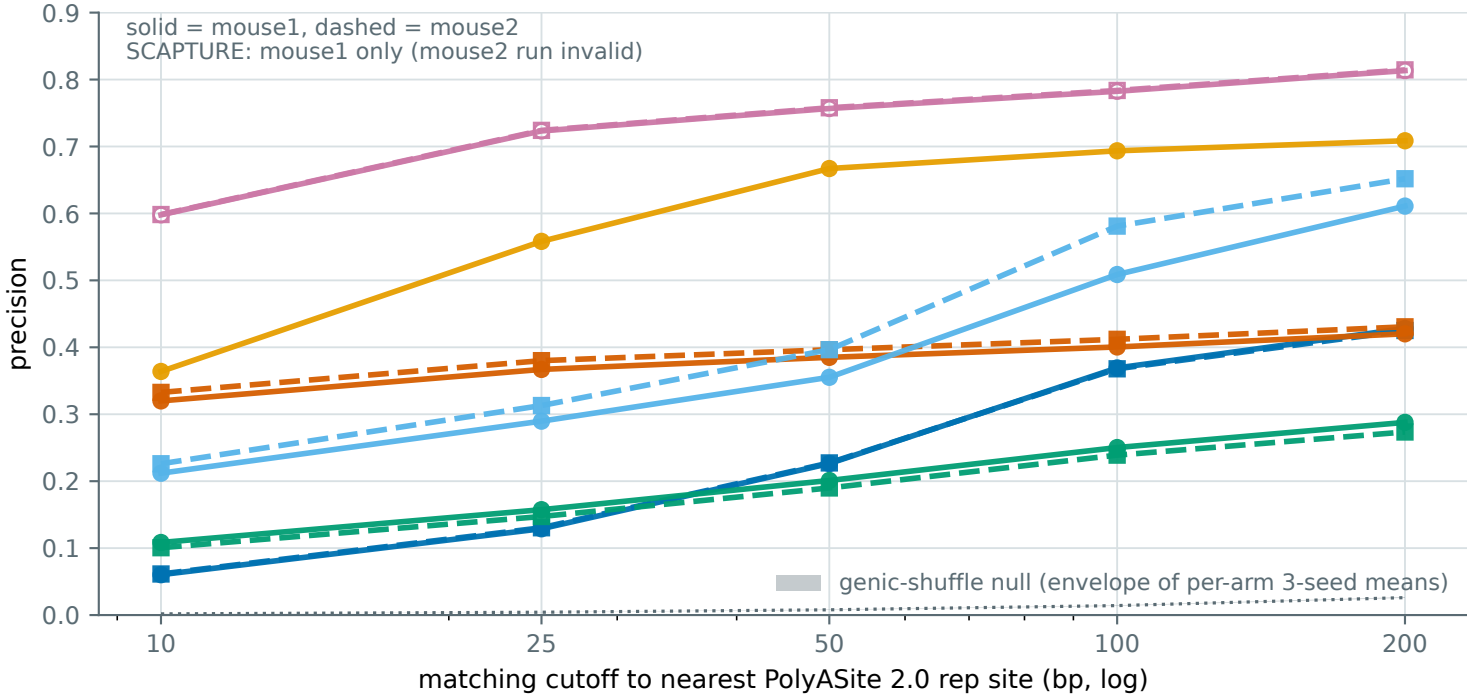
SCAPTURE is the precision arm among the de novo tools on BOTH datasets (P@100 0.65 human / 0.69 mouse) on the fewest-to-middling calls; polyApipe takes the best de novo F1 on both (0.261 human, 0.308-0.312 mouse) but 62% of its mouse calls are depth-1 singletons that reproduce across biological replicates only 0.31 of the time, against 0.81 for its deeper calls; the fewest-calls arm is Sierra on mouse and SCAPTURE on human, and both buy precision that way; scAPATrap calls the most and is least precise. Every rank order changes between the two datasets — no single accuracy number describes any tool in this field.

colour = tool, assigned in alphabetical order so that no tool is visually privileged
PeakATail polyApipe scAPATrap SCAPTURE scUTRquant Sierra

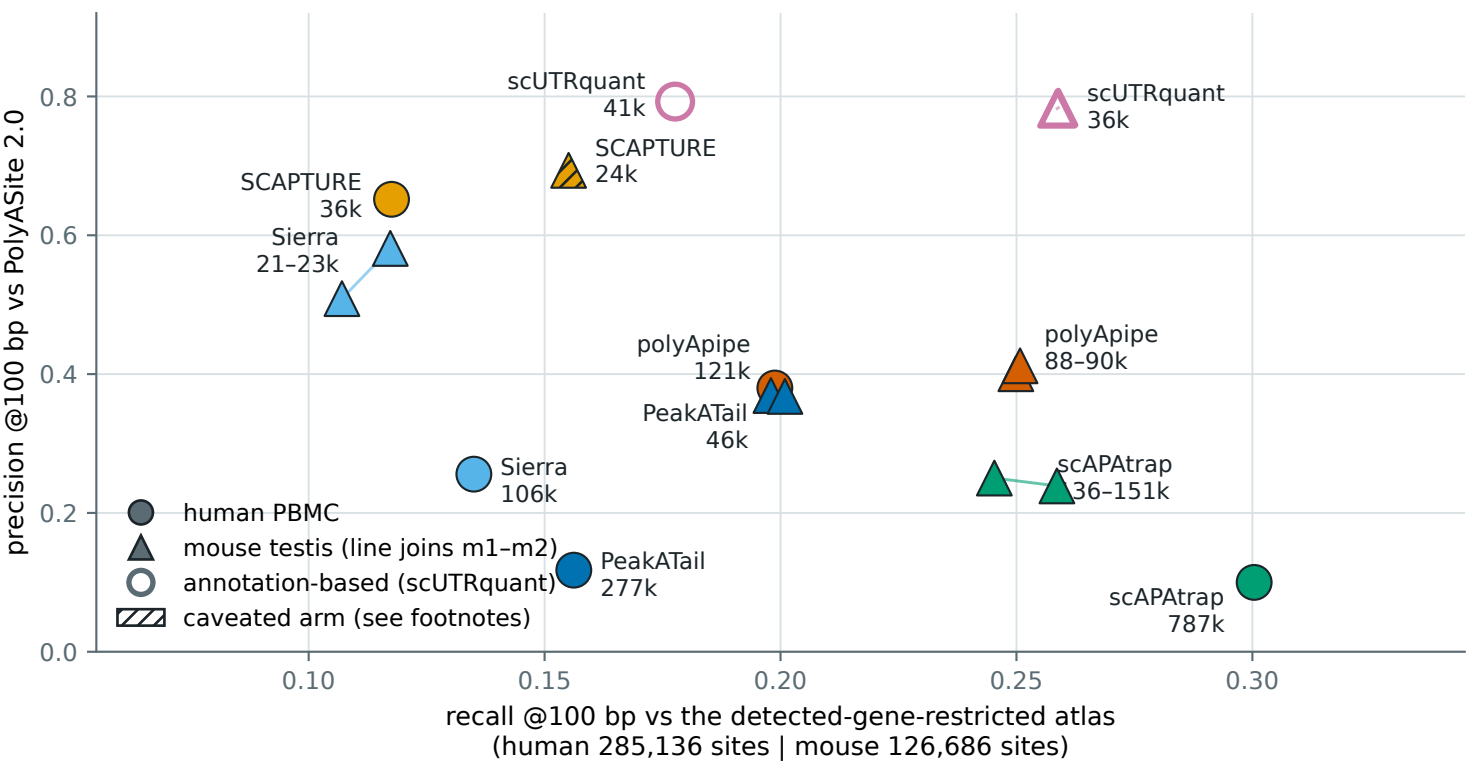
(a1) precision vs cutoff — human PBMC (10x pbmc_10k_v3)



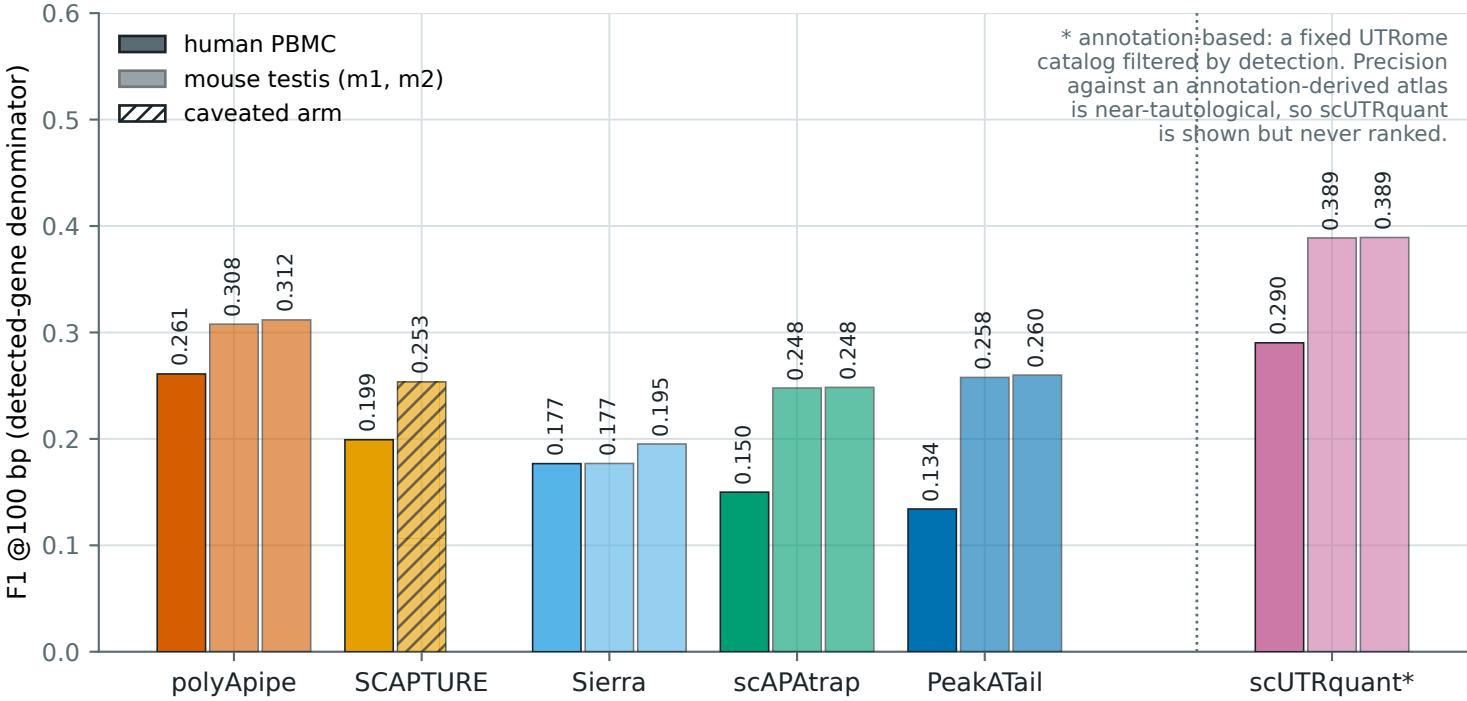
(a2) precision vs cutoff — mouse testis (GSE104556)



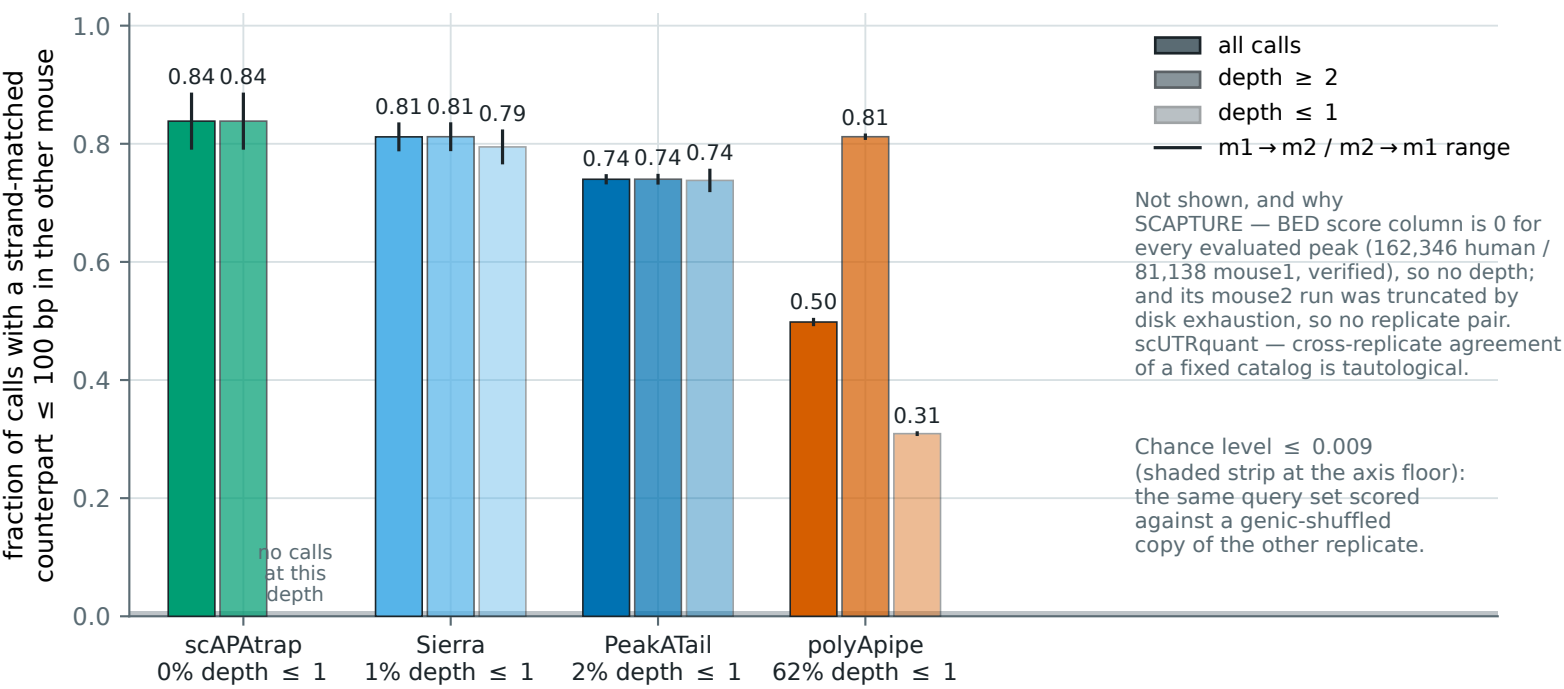
(b) the trade surface: precision-recall @100 bp, labelled with n called



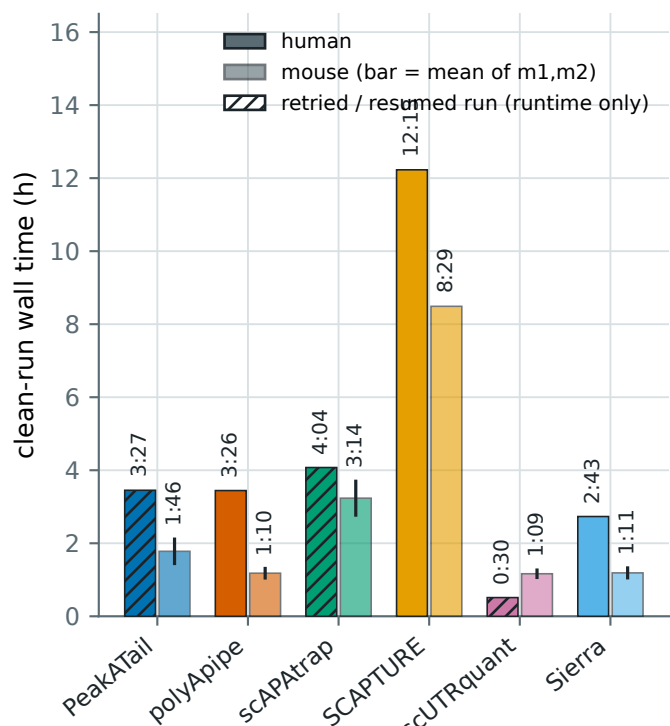
(c) F1 @100 bp — de novo tools ranked by PBMC F1; scUTRquant* separated, not ranked



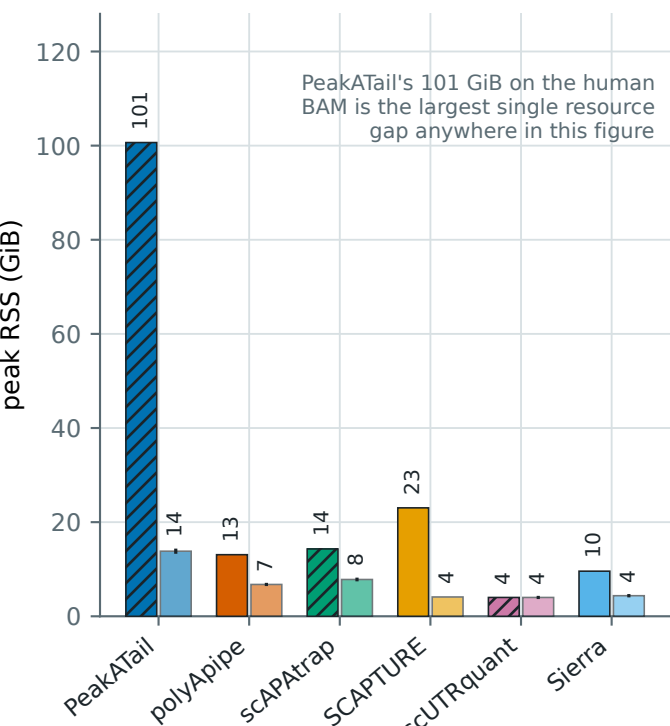
(d) replicate reproducibility, mouse testis — and the call depth it rests on



(e1) wall time



(e2) peak memory



SCORING — One path for every arm (scripts/benchmark_tools/score_tool.py): each tool's output reduced to 1-bp inferred cleavage points, strand-matched with `bedtools shuffle` inside merged annotated gene bodies. Full harvest: benchmark_consolidated.tsv; every plotted value: benchmark_headtohead_results.md. manuscript/09_headtohead_results.md.

CAVEATED ARMS — SCAPTURE mouse (hatched in b, c) is mouse1 only: the mouse2 run was truncated by disk exhaustion and SCAPTURE exited 0 regardless, so no mouse replicate pair exists and its mouse numbers are single-replicate. Hatching in panel (e) marks RUNTIME caveats only, never accuracy: PeakATail human is attempt 4 of 4 (attempts 1-3 ran 1:33:21, 1:52:42 and 4:52:58 before dying on three separate CellRanger-input bugs; those are disclosed, not summed into the bar); scAPATrap human is a resumed run (4:04:29) after attempt 1 was OOM-killed at 8:54:21, and it skipped three already-finished stages, so its bar UNDERSTATES a from-scratch run (12:58:50 of machine time in total); scUTRquant human needed a 4:53 salvage rerun after a missing-nbformat failure, and scUTRquant mouse wall time is dominated by a STARsolo->CellRanger-tag BAM conversion this benchmark imposed (64 and 54 min of the 76 and 64 min totals), not by the pipeline itself.

PANEL (d) — Depth is per-tool and NOT interconvertible: polyApipe = poly(A)-read peakdepth from its GFF, PeakATail = total UMIs (annotated_matrix.mtx row sums), Sierra = total UMIs (CountPeaks matrix row sums), scAPATrap = total UMIs (counts.tsv.gz). scAPATrap has no depth<=1 bar because its own reducePeaks(min.cells=10, min.count=10) step removes such peaks before it reports them, so its high concordance is measured on an already depth-filtered set and is not comparable to the others at face value. PeakATail's depth<=1 stratum includes 715/45,921 (mouse1) and 555/46,672 (mouse2) sites carrying zero UMIs in the filtered-cell matrix. The two directions differ only through their denominators; the vertical line is their range.

READ WITH CARE — scUTRquant is annotation-based and its numbers are near-tautological; it is shown for context and never ranked with the de novo tools. n called spans 21k-787k (37-fold), so precision and recall must always be read together with panel (b). SCAPTURE is annotation-GUIDED (peaks called inside gene models, then filtered by the DeepPASS sequence classifier), the likeliest source of its precision lead. PeakATail's PBMC deficit is not a threshold artefact (its top 22k / 36k / 106k calls by UMI give precision 0.119 / 0.120 / 0.118, identical to the full set) and not a coordinate offset (relaxing to 200 bp lifts it only 0.118 -> 0.166 while polyApipe goes 0.380 -> 0.410).